Can memistor-class electrochemical weights train neural networks?

A pre-registered simulation study from Widrow's 1960 memistor to modern ECRAM, with a device specification sheet and an experimental roadmap

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October 2026

Abstract

In 1960 Widrow and Hoff stored the weights of their adaptive neuron in memistors: three-terminal electrochemical cells whose conductance is set by the charge through a third terminal. Their modern descendants are lithium-ion, metal-oxide and protonic ECRAM. We ask, in simulation, whether such devices can train neural networks in memory to near-digital accuracy, what specifications they need, and what energy advantage survives once the periphery is counted.

Simulator and validation. Our device model includes levels, asymmetry, saturation, write noise, spreads, drift and reference error. Every check is shown to fail on a planted fault. We validate it against a published analog-training result: for the RPU-baseline device on MNIST we obtain 1.78%, 14.47% and 2.26% test error for the exact update, SGD and Tiki-Taka. The published values are 2.0%, about 15% and about 2%. We also reproduce the published robustness of chopped TTv2 and AGAD to reference offsets.

Main comparison. In a pre-registered comparison with five seeds per arm, we test four published device specifications and Widrow's memistor on MNIST, a CIFAR-10 subset and a character language model on human-written text.

- Plain SGD fails on every published device and task.
- With a Tiki-Taka-family algorithm, the Li-ECRAM and protonic specifications reach the baseline within our gap lines or within five-seed resolution on all three tasks.
- The best language-model gap is -1.29% [-4.46, 1.87].

Scale. When every arm tunes its own learning rate, the gap does not grow with width; an earlier observation of growth was a learning-rate artefact.

Specification sheet. Sensitivity sweeps show that the binding specifications are an accurate symmetry-point reference ($\sigma_r \leq 0.02$ of the half-range for TTv2) and a saturating response. A few hundred levels suffice.

Energy. With periphery, at the efficiency of a measured analog-AI chip, training energy per weight is about 32× below an H100 at its measured utilisation, 15–234× over our sensitivity range. The read periphery, not the femtojoule device, dominates.

We release the simulator, the records, a device-experiment roadmap with pass criteria, and a bench replication of the 1960 memistor.

1 Introduction

Training a neural network on a digital processor moves every weight from memory to arithmetic and back at every update. In 1960, Widrow and Hoff took a different route: they stored each weight of their adaptive ADALINE neuron in a *memistor*, an electrochemical cell whose conductance is set by the time integral of a current through a third terminal [1–3]. The weight was read without

being disturbed, and it was adapted in place. The device was macroscopic, the cells were wet, and no multilayer learning rule existed yet, so the line ended with the Madaline hardware of the early 1960s [3]. That machine was a working analog learning system.

Sixty years later, three-terminal electrochemical devices have returned at the nanoscale. Lithium-ion ECRAM [4, 5], CMOS-compatible metal-oxide ECRAM [6], protonic programmable resistors [7, 8] and open-loop programmable arrays [9] all decouple the read path from the write path, just as the memistor did. They report hundreds to thousands of conductance levels, nanosecond pulses and femtojoule-scale write energies. In parallel, algorithms for training on imperfect analog weights have matured: the stochastic pulse-coincidence update of the resistive processing unit (RPU) [10], Tiki-Taka [11] and its successors TTv2 [12], chopped TTv2 and AGAD [13], and residual learning with convergence guarantees [14].

This paper asks whether memistor-class three-terminal electrochemical weights, from Widrow's cell to today's ECRAM, can support analog in-memory *training* at near-digital accuracy. If they can, what device specifications does that need, and what energy advantage remains once the periphery is counted? We could not build devices. Instead we provide what groups that can build them need to test the idea quickly:

1. **A validated simulator.** We reproduce a published analog-training result (Gokmen and Haensch 2020) with an independent device model, and we qualitatively check the offset-robustness trend of Rasch et al. 2024 (Section 4.1).

2. **A pre-registered comparison** of plain SGD and three device-tolerant algorithms on four published electrochemical device specifications and Widrow's memistor. There are three tasks (MNIST, a CIFAR-10 subset, and a character-level language model on human-written text) and five seeds, with confidence intervals (Section 4.2).

3. **A scale study** in which every arm gets its own learning rate at every width. It tests an earlier observation that the device gap grows with model width (Section 4.3).

4. **A device specification sheet** built from one-at-a-time and pairwise sensitivity sweeps: the minimum levels, and the maximum asymmetry, nonlinearity, write noise, drift and reference error for a given accuracy gap at each model size (Section 4.4).

5. **A system-level energy model** with periphery, built only from published figures and our simulated pulse counts. It compares against an H100 at its measured utilisation and separates what is measured from what is projected (Section 5).

6. **A roadmap** of device experiments with pass criteria, array sizes, a hardware-in-the-loop protocol, and a bench replication of the 1960 memistor with safety notes (Section 7, Appendix A).

Every experiment was pre-registered before it ran. The plan, every deviation, the simulator, its checks (each shown to fail on a planted fault), every run record and the scripts that turn records into the numbers and figures of this paper are released together (Section 9). All results are simulations, and every device parameter carries a provenance label (measured, fitted, claimed, estimated or assumed).

2 Background and related work

2.1 The memistor (1960)

Widrow and Hoff introduced the ADALINE and the least-mean-squares (LMS) rule in 1960 [2]. To store its weights they built the memistor [1]. A pencil-lead (graphite) substrate sits in a copper-sulphate plating bath. Copper is plated onto it or removed from it by a current through a third electrode, and the conductance between the lead's two ends is sensed with a small AC signal that does not disturb the state. Widrow, Pierce and Angell described elements with "thousands" of analog levels that spanned roughly 100 to 1 ohm in about ten seconds at a few milliamperes; this description is reported in secondary sources, see [15]. Mouttet analyses the memistor as a three-terminal memristive system and summarises Widrow's ideal: a conductance linear in the plating charge, reversible, non-volatile and free of hysteresis [15]. Widrow and Lehr recount that the Madaline I machine was built in hardware with memistor weights and used for pattern recognition research [3]. Write noise, up/down asymmetry, retention and endurance were never reported in a form we could find, so in our catalog these are estimates (Section 3.4). The memistor predates Chua's memristor [16] and its later physical identification in titanium-oxide films [17]. It is the conceptual ancestor of today's three-terminal synaptic transistors.

2.2 Device requirements for training

Gokmen and Vlasov [10] defined the resistive processing unit (RPU). Updates are stochastic pulse trains whose coincidences change the conductance by a fixed step (stochastic computing, [18]). They derived device specifications for plain SGD on MNIST at a small error penalty: about a thousand levels, up/down asymmetry of a few percent, and tolerance to large cycle-to-cycle and device-to-device variation. Agarwal et al. [19] reached similar conclusions; the requirements were extended to convolutional networks [20], and reviews cover the device landscape [21–24]. Mixed-precision schemes [25] accumulate the gradient digitally and program the devices rarely. They relax the device demands but give up the fully parallel in-memory update.

2.3 Algorithms that tolerate asymmetric devices

Real devices change their conductance by different amounts up and down, and the step depends on the state (soft bounds). Under plain SGD this acts as an implicit penalty that pulls weights toward each device's symmetry point [11, 14].

- **Tiki-Taka (TTv1)** [11] uses two arrays. The gradient is accumulated on an array A, and its columns are periodically transferred into the weight array C, with $W = \gamma A + C$. Asymmetry then acts as a useful decay on A instead of a bias on W.
- **TTv2** [12] inserts a digital buffer H between the two arrays. H low-pass filters the reads of A, and a pulse is sent to C only when a threshold is crossed. TTv2 still needs every device's symmetry point to be subtracted by a reference.
- **Chopped TTv2 (c-TTv2)** and **AGAD** [13, 26] remove this sensitivity. A random sign modulates the inputs and the reads (c-TTv2). AGAD additionally replaces the static reference with a digital, on-the-fly reference taken from the recent history of A.
- **Residual learning** [14] gives convergence guarantees for this family.

IBM's open-source AIHWKIT [27] implements these algorithms with fitted device presets, including the Li-ECRAM and metal-oxide ECRAM models we use.

2.4 Electrochemical three-terminal devices

- **Lithium-ion synaptic transistors and ECRAM.** Fuller et al. [5] showed a lithium-ion synaptic transistor, followed by an ionic floating-gate array [28]. Tang et al. [4] reported Li-WO_3 ECRAM with about a thousand symmetric, linear states and nanosecond pulses.
- **Metal-oxide ECRAM.** Kim et al. [6] made it CMOS-compatible.
- **Protonic resistors.** Onen et al. [7] reported nanosecond protonic programmable resistors with a phosphosilicate-glass electrolyte. In 2026, Xu et al. [8] quantified cycle-to-cycle (3%) and device-to-device (21%) variability of BEOL-compatible protonic ECRAM.
- **Open-loop programmable arrays.** Chen et al. [9] demonstrated a 10×10 W/YSZ/WO_x array with 256 states.
- **Organic devices.** Organic electrochemical synapses [29] extend the family to low voltages.

In-situ training on ECRAM hardware has so far been shown on small arrays, for example a 3×3 array by Li et al. [30] and the 10×10 array of [9]. Other memory technologies have gone further. Phase-change and resistive arrays have trained or run networks at larger scale [31–34], and analog-AI inference chips with on-chip periphery now report measured efficiencies of about 10 TOPS/W [35, 36].

3 Methods

3.1 Device model

Each analog weight is one device with a normalised state $u \in [-1, 1]$ between its own bounds. Its conductance is $g = g_{\min} + \frac{u+1}{2}(g_{\max} - g_{\min})$, and the weight is read differentially as $w = \alpha\,(g - g_{\text{ref}})$. One nominal pulse moves u by $k = 2/n_{\text{states}}$ at mid-range. With asymmetry a the up and down steps are $k_\pm = k\,s\,(1 \pm a/2)$, where s is a per-device step factor with spread σ_{d2d}. With saturation (soft bounds) the steps depend on the state:

$$\Delta u_+ = k_+\,(1 - \gamma_+ u), \qquad \Delta u_- = -k_-\,(1 + \gamma_- u). \tag{1}$$

The per-device slopes $\gamma_\pm$ have a relative spread σ_γ. A device's symmetry point, where an up and a down pulse cancel, is $u^* = (k_+ - k_-)/(k_+\gamma_+ + k_-\gamma_-)$. The reference g_{ref} is set at $g(u^*)$ plus a calibration error of standard deviation σ_r, in units of the half-range. A constant-step device has no symmetry point and is referenced at mid-range.

- **Several pulses in one update.** When a device receives n_+ up and n_- down pulses in one update, the pulses are taken as uniformly interleaved, and $\mathrm{d}u/\mathrm{d}s = A - Bu$ is solved exactly over the batch, with $A = k_+ n_+ - k_- n_-$ and $B = k_+\gamma_+ n_+ + k_-\gamma_- n_-$.
- **Write noise.** Cycle-to-cycle noise adds a Gaussian with standard deviation c_{2c} times the local step, growing as the square root of the pulse count.
- **Drift.** Retention loss relaxes u toward the bare state $u = -1$ with a time constant τ in training steps.
- **Bounds.** The bounds $g_{\min}$, $g_{\max}$ carry a device-to-device spread.

Forward and backward passes read the weights ideally, in fp32. Read noise and DAC/ADC quantisation are outside the scope of this study, which isolates the *update*. Biases, layer norms and the language model's position table are digital.

3.2 The update

We use the stochastic pulse-coincidence scheme of [10]. Each sample (row) offers $BL = 31$ pulse slots. Input i fires with probability $p_{x,i} \propto |x_i|$ and error j with probability $p_{d,j} \propto |d_j|$, with the scaling split equally between the two streams (update management). Weight ij therefore receives coincidences with mean $\eta\,|x_i d_j|/(\alpha k)$. Up and down counts are kept separate, so asymmetry is never cancelled digitally. Within a batch the counts of all samples are summed and drawn from a Poisson distribution with the exact mean. A mini-batch therefore only groups pulses; it does not average gradients.

3.3 Algorithms

SGD. The pulses go directly into the weight devices.

TTv1 [11]. The pulses go into an array A, and the forward weight is $C + \gamma A$. Every batch, n_c columns j of A are read and added to C by a pulsed outer product, $\lambda A_{:,j}\, e_j^\top$. A is never reset.

TTv2 [12]. Each read column is added to a digital buffer, $H_{:,j} \mathrel{+}= \lambda A_{:,j}/(\alpha k)$. C receives up to 31 pulses where $|H| \geq 1$, and H keeps the remainder. The read column of A is reset to its symmetry point, landing on the device's own zero, which is off by the reference error.

c-TTv2 [13]. Chopper signs $c_j \in \{\pm 1\}$ modulate the inputs of the A update ($c \odot x$) and demodulate the reads ($c_j A_{:,j}$). Each read column's chopper flips with probability ρ.

AGAD [13]. The choppers of c-TTv2, plus a digital leaky average of the reads, $p_j \leftarrow (1-\beta)p_j + \beta A_{:,j}$. The read is $z_j = c_j(A_{:,j} - p_j^{\mathrm{ref}})$, and p_j^{ref} is reset to p_j whenever c_j flips.

In every TT-family algorithm the A array uses a learning rate $1/\lambda$ times larger, so that $\lambda\times$ (the A rate) equals the SGD rate. The settings were chosen once on MNIST validation with a mid-quality device that is not in the catalog (Section 3.6; Appendix B). We implemented the algorithms from the papers' text; we did not use the authors' code.

3.4 Device catalog and provenance

Table 1 lists the devices. Every field carries a provenance label:

- MEASURED: reported in the paper;
- FIT: from the AIHWKIT presets fitted to the published devices [27];
- CLAIM: a qualitative statement in the paper, translated by us;
- ESTIMATE: ours;
- ASSUMED: for example, the 2% reference error given to every device.

The protonic device of [7] is mostly estimated: the paper reports a thousand up and a thousand down pulses that are "nearly linear and symmetric", but no write-noise or variability statistics. Widrow's memistor is given two specifications, one as he described it and one pessimistic, together with a pencil-lead cell built today with modern hobby materials (Appendix A).

Table 1: Device catalog (simulator units; Section 3.1). Levels n_{states}, global asymmetry a, saturation γ_+/γ_-, write noise c_{2c} (per pulse, relative to the step), step spread σ_{d2d}, drift time constant τ (training steps), reference error σ_r. Provenance per field is in records/CATALOG.json.

Device	Label	n_{st}	a	γ_+/γ_-	c_{2c}	σ_{d2d}	τ	σ_r
Li-ECRAM (Tang 2018)	PUBLISHED-FIT	1000	0	0.12/0.51	0.3	0.1	none	0.02
MO-ECRAM (Kim 2019)	PUBLISHED-FIT	7089	0	0.42/0.73	2	0.1	none	0.02
Protonic (Onen 2022)	PARTLY-PUBLISHED	1000	0.05	0.20/0.20	0.3	0.1	none	0.02
W/YSZ/WO_x (Chen 2023)	PUBLISHED-FIT	256	0	0.63/0.63	2.048	0.023	none	0.02
Memistor (Widrow 1960, as claimed)	CLAIM+ESTIMATE	1000	0.05	0.10/0.10	0.3	0.2	none	0.02
Memistor (pessimistic)	ESTIMATE	100	0.2	0.30/0.30	1	0.3	1000	0.05
Pencil lead built today	ESTIMATE	100	0.1	0.30/0.30	0.5	0.2	940	0.05

3.5 Tasks and baselines

MNIST [37]. A 784–256–128–10 tanh MLP; 10 epochs, batch 128, on 55,000 training images. 5,000 held-out training images form the validation set, and the 10,000 test images are the test set.

CIFAR-10 subset [38]. Three 3×3 convolutions with c, $2c$ and $2c$ channels (base $c = 32$), computed as crossbar matrix products over unfolded patches, then two fully connected layers. 30 epochs, batch 128, on the first 10,000 images of training batches 1–2. The next 2,000 images are the validation set, and the 10,000-image test set is the test set.

Character LM. A byte-level, two-layer transformer [39] of width d (base 128) and context 128, trained on 8 MB of enwik8 [40], which is human-written Wikipedia text of 2006. 3,000 steps of 32×128 bytes. Every matrix is analog, including the embedding and the head. Validation is the slice's last 0.5 MB, held out from training in selection runs. Test is a separate 0.5 MB slice.

The baseline ("FP") is the same network, initialisation, data order and update rule with an exact floating-point weight update. Adam [41] on digital twins is reported for the LM as context only. The gap is FP accuracy minus device accuracy for classification, in percentage points. For the LM it is the device's bits per byte relative to FP's, minus one, in percent.

3.6 Protocol and statistics

Every experiment was pre-registered in a committed plan before it ran (Section 9).

- **Hyperparameters** were chosen on validation splits only, with an edge rule: whenever the best value of a grid lies at its edge, the next half-decade point is added. The learning rate of every device arm in the main table is the FP validation choice. The scale study instead tunes each arm separately.
- **Statistics.** The main table uses five seeds per arm. Gaps are paired by seed, and we report the mean and a 95% Student-t interval. A cell PASSES when the whole interval lies below the gap line (2 pp for classification, 5% for the LM), FAILS when it lies wholly above, and is UNRESOLVED otherwise.
- **Multiplicity.** We make no correction for multiplicity. The intervals describe each cell, not the family.

3.7 Simulator checks

The simulator is checked by 14 checks, and each check is shown to fail on at least one deliberately planted fault (17 faults in all). Examples of what the checks cover:

- the exact-update arm equals PyTorch SGD;
- the mean pulse update equals the SGD step;
- alternating pulses converge to the analytic symmetry point, including with per-device slopes;
- write noise grows as $\sqrt{n}$;
- drift relaxes to the bare state;
- a static read offset is integrated by TTv2, mostly cancelled by c-TTv2 and nearly fully cancelled by AGAD;
- TTv1 moves C along $+\lambda A$;
- the SGD and TTv2 paths are bit-identical to the earlier code that produced our preliminary results.

The planted faults include a swapped pulse sign, a dropped chopper demodulation and a never-updated AGAD reference. Both suites run at the start of every compute session and stop it on any failure (Appendix C).

4 Results

4.1 Simulator validation

Gokmen and Haensch 2020. We map their RPU-baseline device onto our model:

- step 10^{-3} at $w = 0$;
- mean saturation slope 1.66 with a 25% per-device spread;
- 30% step spread and 30% cycle-to-cycle noise;
- a symmetry point at $w = 0$ for every device.

We train their 784–256–128–10 sigmoid network on MNIST for 50 epochs at their learning rate of 0.01. The results are in Table 2 and Fig. 1a.

- FP reaches 1.78% test error, against their 2.0%.
- SGD on the device reaches 14.47% [12.68, 16.26], against "about 15%".
- Tiki-Taka reaches 2.26% [2.13, 2.39], against "close to 2%".

By our pre-registered criteria the agreement is **quantitative**. The known differences are listed in Section 6.1:

- we use mini-batches of 16, not 1;
- we do not model read noise, ADC/DAC quantisation or analog biases.

At batch 64 the FP and TTv1 results barely move, but SGD on the device degrades further. Summing more samples' pulses before the weight changes hurts a strongly asymmetric device, which is another reason to use batch 16 as the closer match to their batch 1.

Table 2: Validation against Gokmen and Haensch 2020 (FCN 784-256-128-10, MNIST, 50 epochs): test error in %, our mean and 95% interval over 3 seeds at batch 16, and one seed at batch 64.

Arm	Published [11]	Ours (batch 16)	Ours (batch 64)
FP (exact update)	2.0	1.78 [1.75, 1.81]	1.79 (n=1)
SGD on the RPU-baseline device	about 15	14.47 [12.68, 16.26]	24.32 (n=1)
Tiki-Taka (TTv1)	close to 2	2.26 [2.13, 2.39]	2.07 (n=1)
TTv2 (ours, 16 columns per batch)	–	2.59 [2.42, 2.75]	not run

Table 3: Qualitative check of Rasch et al. 2024 (Fig. 5): a 20-state soft-bounds device with an uncorrected reference offset σ_r; test error in % (mean and range over 2 seeds as a t-interval; 10 epochs). FP: 2.69 [0.47, 4.92].

Offset	TTv2	c-TTv2	AGAD
$\sigma_r = 0$	4.12 [2.79, 5.46]	4.29 [3.91, 4.67]	8.47 [7.58, 9.36]
$\sigma_r = 0.5$	24.27 [18.23, 30.30]	13.02 [10.99, 15.05]	7.88 [3.49, 12.26]

Rasch et al. 2024, qualitatively. We use a 20-state soft-bounds device with an uncorrected reference offset of $\sigma_r = 0.5$ (Table 3, Fig. 1b).

- TTv2's test error rises from 4.12% to 24.27%.
- c-TTv2 rises less, from 4.29% to 13.02%.
- AGAD is unaffected: 8.47% versus 7.88%.

This is the published ordering; our pre-registered check reads **reproduced**. In our implementation AGAD starts from a higher error at zero offset on this very coarse device. Its dynamic reference costs accuracy when no offset needs correcting, a trade-off we see again in Section 4.2.

Not run. A pre-registered optional cross-check of a few cells against IBM's AIHWKIT [27] was the lowest item in our priority list and was not run. The quantitative agreement with a published result stands in its place. A cross-check is a natural first step for anyone extending this work.

4.2 Main comparison: four published devices and the memistor

Table 4 and Fig. 2 give the gap to the exact-update baseline over five seeds.

Plain SGD fails. On every published electrochemical device, plain SGD lies wholly above the line on all three tasks (4, 4 and 4 of four devices FAIL on MNIST, CIFAR and the LM). This holds even though the Li-ECRAM, protonic and metal-oxide specifications have about a thousand or more levels. Their asymmetric, saturating steps pull the weights toward their symmetry points, as predicted by [11, 14].

The Tiki-Taka family recovers most of the gap.

- **Best cell per task** among the published devices:
  - MNIST: MO-ECRAM (Kim 2019) with AGAD, 0.49 pp [0.33, 0.66], PASS;

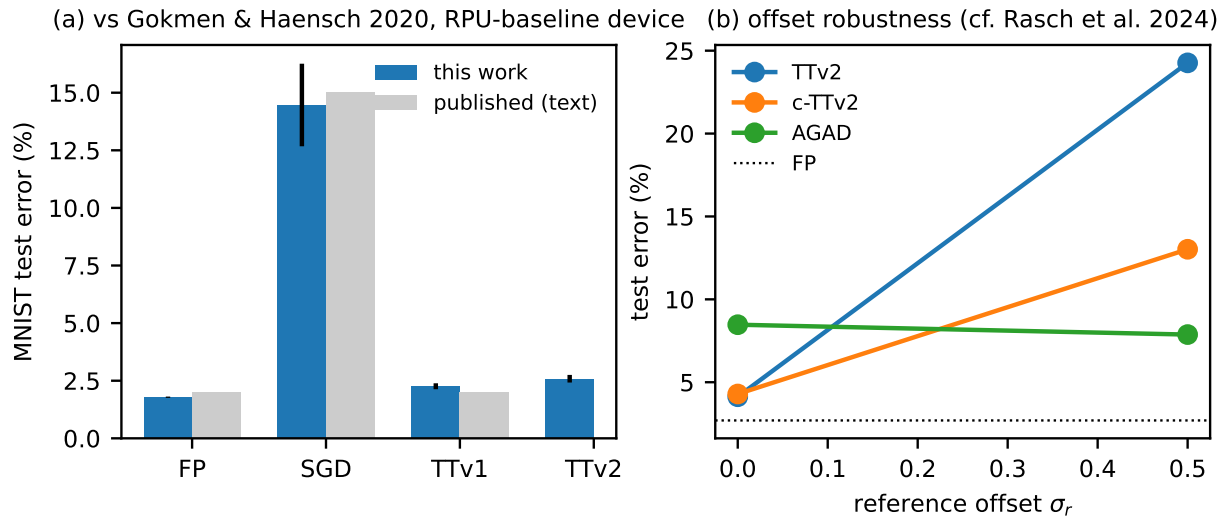


Figure 1: Simulator validation. (a) Test error of our simulator (mean and 95% interval, 3 seeds) against the values stated in the text of Gokmen and Haensch 2020 for their FCN on MNIST with the RPU-baseline device. (b) Robustness to an uncorrected reference offset on a 20-state device (2 seeds), the qualitative trend of Rasch et al. 2024, Fig. 5.

  - CIFAR: Protonic (Onen 2022) with c-TTv2, 0.08 pp [-2.57, 2.73], UNRESOLVED;
  - LM: Li-ECRAM (Tang 2018) with c-TTv2, -1.29% [-4.46, 1.87], PASS.

- **Strictest reading** (whole interval below the line on every task):
  - Mixing algorithms across tasks, 2 devices pass all three tasks: Li-ECRAM (Tang 2018) and Protonic (Onen 2022).
  - With one algorithm on all three tasks, 0 pairings pass everywhere.

- **No cell fails.** 8 device–algorithm pairings have no FAIL on any task: Li-ECRAM (Tang 2018) with TTv2; Li-ECRAM (Tang 2018) with c-TTv2; Li-ECRAM (Tang 2018) with AGAD; Protonic (Onen 2022) with TTv2; Protonic (Onen 2022) with c-TTv2; Protonic (Onen 2022) with AGAD; Memistor (Widrow 1960, as claimed) with c-TTv2; Memistor (Widrow 1960, as claimed) with AGAD. Their remaining cells are UNRESOLVED: the interval contains the line.

The limiting factor is statistical as well as physical. On the CIFAR subset the FP baseline itself varies by 1.41 pp (standard deviation over seeds), so five seeds give intervals about ±2 pp wide.

The devices separate clearly.

- **Li-ECRAM and protonic** are within or near the lines.
- **Metal-oxide ECRAM** is good on MNIST but fails on the LM. Its fitted write noise is twice its (very small) step.
- **W/YSZ/WO$_x$** fails on CIFAR and the LM with every algorithm. Its 256 states and strong nonlinearity are not enough for these tasks, although they suffice for its published 10×10 demonstration.

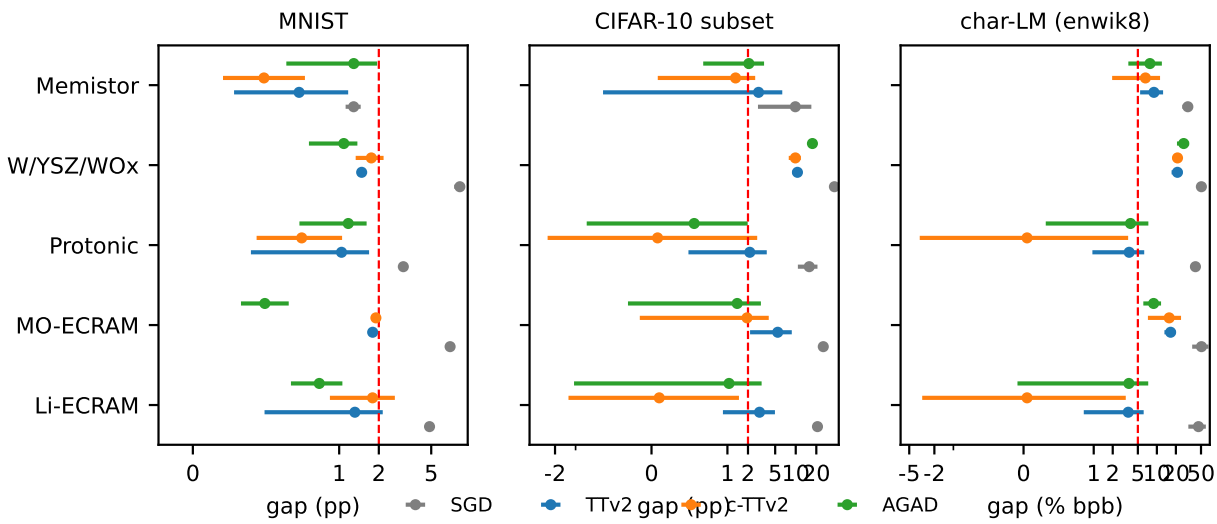


Figure 2: Gap to the exact-update baseline (mean and 95% interval over 5 seeds; symmetric-log axis); the dashed line is the pre-registered gap line.

- **Widrow's memistor, as he described it** (thousands of levels, near-linear), does as well as the modern devices on MNIST, where even plain SGD passes on it because his description implies an almost symmetric device. On the LM it sits between the good and the poor modern devices.

The memistor's ideal was right; what the 1960 device lacked was density, speed and energy, not accuracy (Section 5).

Among the tolerant algorithms.

- c-TTv2 gives the best LM results on Li-ECRAM and protonic devices.
- AGAD is the most uniform on MNIST, but it is not uniformly better. On the noisiest device (metal-oxide) AGAD is the best tolerant algorithm on the LM, while on the coarse 20-state device of Section 4.1 it pays an accuracy cost.
- Plain TTv2, our pre-registered choice for the specification sheet because it scored best on validation, is not the best on the LM for any device except, marginally, W/YSZ/WO$_x$, where every algorithm fails.

Against Adam. The pre-registered baseline is SGD with an exact update. For context, the FP-SGD language model is 23.3% worse in bits per byte than the same network trained by Adam in digital. The best analog cells are about as far from Adam as FP-SGD is. Analog training here inherits SGD's optimiser gap; closing that gap needs analog-friendly adaptive methods, which are beyond this study.

4.3 Scale: the gap does not grow when every arm has its own learning rate

An earlier exploratory screen of ours (Appendix D) found that the device gap grows with width once the FP learning rate is re-tuned per width. If true, that would be a serious obstacle, so we tested it directly.

Table 4: Main result (E3): gap to the exact-update baseline, mean and 95% t-interval over 5 seeds, paired by seed. MNIST and CIFAR in percentage points of test accuracy; LM in % of bits per byte. P = the whole interval is below the line (2 pp; 5%), F = wholly above, U = the interval contains the line. Every arm uses the FP validation learning rate.

Device	Algorithm	MNIST gap (pp)	CIFAR-10 subset gap (pp)	char-LM gap (%)
FP baseline (acc. %; bpb)		97.85 [97.79, 97.92]	60.39 [58.64, 62.14]	2.912 [2.822, 3.001]
Li-ECRAM (Tang 2018)	SGD	4.85 [4.73, 4.98] F	20.81 [18.06, 23.57] F	45.58 [31.52, 59.64] F
	TTv2	1.32 [0.49, 2.14] U	2.95 [0.94, 4.96] U	3.52 [0.86, 6.18] U
	c-TTv2	1.79 [0.94, 2.65] U	0.10 [-1.27, 1.47] P	-1.29 [-4.46, 1.87] P
	AGAD	0.86 [0.67, 1.06] P	1.06 [-1.05, 3.17] U	3.60 [-0.09, 7.30] U
MO-ECRAM (Kim 2019)	SGD	6.96 [6.60, 7.31] F	25.29 [22.15, 28.44] F	50.84 [36.24, 65.45] F
	TTv2	1.80 [1.68, 1.91] P	5.45 [2.15, 8.75] F	16.51 [13.17, 19.84] F
	c-TTv2	1.90 [1.81, 1.99] P	1.94 [-0.15, 4.03] U	15.68 [7.21, 24.14] F
	AGAD	0.49 [0.33, 0.66] P	1.39 [-0.31, 3.09] U	8.88 [6.10, 11.65] F
Protonic (Onen 2022)	SGD	3.07 [2.85, 3.29] F	15.77 [10.75, 20.79] F	40.82 [34.31, 47.33] F
	TTv2	1.04 [0.40, 1.69] P	2.13 [0.49, 3.77] U	3.64 [0.98, 6.30] U
	c-TTv2	0.74 [0.44, 1.05] P	0.08 [-2.57, 2.73] U	-0.84 [-4.30, 2.62] P
	AGAD	1.17 [0.73, 1.62] P	0.56 [-0.85, 1.98] P	3.82 [0.32, 7.33] U
W/YSZ/WO_x (Chen 2023)	SGD	8.25 [7.47, 9.02] F	36.82 [33.17, 40.47] F	50.67 [47.53, 53.81] F
	TTv2	1.48 [1.37, 1.60] P	10.59 [8.94, 12.24] F	21.17 [17.11, 25.24] F
	c-TTv2	1.75 [1.33, 2.17] U	9.87 [7.90, 11.84] F	21.23 [17.81, 24.66] F
	AGAD	1.08 [0.79, 1.37] P	17.55 [15.52, 19.58] F	26.59 [20.81, 32.38] F
Memistor (Widrow 1960, as claimed)	SGD	1.29 [1.12, 1.46] P	9.86 [2.80, 16.92] F	30.94 [25.31, 36.57] F
	TTv2	0.73 [0.28, 1.17] P	2.86 [-0.64, 6.35] U	8.99 [5.43, 12.54] F
	c-TTv2	0.49 [0.21, 0.77] P	1.32 [0.08, 2.55] U	6.59 [1.96, 11.23] U
	AGAD	1.29 [0.64, 1.94] P	2.06 [0.68, 3.44] U	7.76 [3.53, 11.99] U

Design.

- Widths: CIFAR base channels $c \in \{16, 32, 64\}$ and LM widths $d \in \{64, 128, 256\}$, with a fixed training budget.
- Arms: FP; and TTv2 and c-TTv2 on the protonic and Li-ECRAM devices.
- Learning rates: every arm chooses its own on validation at every width, with the edge rule.
- Seeds: three test seeds per arm and width.
- Verdict: GROWS, SHRINKS or FLAT, by the pre-registered rule (Section 3.6).

Result (Table 5, Fig. 3).

- Of 8 device–algorithm arms, 0 GROW, 6 are FLAT and 2 SHRINK. The two that shrink are TTv2 on the LM, for both devices.
- The earlier observation is therefore **not confirmed**.
- On the LM the tuned device arms are *better* than FP-SGD, and increasingly so at width 256. The protonic device with c-TTv2 is -12.97% at $d = 256$ and -6.38% at $d = 128$.

Why the earlier screen saw growth.

- The earlier runs, like our specification sweep (Section 4.4), used the *FP* learning rate for the device.

Table 5: Scale (E4): every arm uses its own validation-chosen learning rate (in parentheses) at every width; gap (pp for CIFAR, % for the LM), mean and 95% interval over 3 seeds paired with FP at its own learning rate. Verdict by the pre-registered rule (Section 4.3).

Task	Width	FP metric (lr)	Li-ECRAM c-TTv2	Li-ECRAM TTv2	Protonic c-TTv2	Protonic TTv2
CIFAR	16	55.70 (0.0003)	0.64 [-1.53, 2.82] (0.003)	1.86 [-0.64, 4.35] (0.001)	-2.13 [-7.14, 2.88] (0.003)	0.41 [-1.44, 2.26] (0.001)
CIFAR	32	60.15 (0.001)	-0.40 [-3.13, 2.33] (0.001)	2.12 [-1.66, 5.90] (0.001)	-1.88 [-6.35, 2.59] (0.003)	-0.21 [-5.27, 4.84] (0.003)
CIFAR	64	60.87 (0.0003)	-1.34 [-5.57, 2.89] (0.001)	1.11 [0.17, 2.05] (0.001)	-1.30 [-3.56, 0.96] (0.003)	0.33 [-2.43, 3.09] (0.001)
		verdict:	FLAT	FLAT	FLAT	FLAT
LM	64	3.193 (0.0001)	-0.55 [-8.58, 7.48] (0.0001)	-0.31 [-6.26, 5.63] (0.0003)	-7.21 [-11.77, -2.65] (0.0003)	-3.45 [-6.44, -0.46] (0.0003)
LM	128	2.881 (0.0001)	-0.09 [-7.23, 7.04] (0.0001)	-0.60 [-8.73, 7.53] (0.0003)	-6.38 [-12.53, -0.23] (0.0003)	-3.88 [-10.90, 3.14] (0.0003)
LM	256	3.100 (3e-05)	-13.73 [-14.66, -12.79] (0.0001)	-9.01 [-10.21, -7.82] (0.0001)	-12.97 [-13.57, -12.36] (0.0001)	-8.48 [-9.67, -7.29] (0.0001)
		verdict:	FLAT	SHRINKS	FLAT	SHRINKS

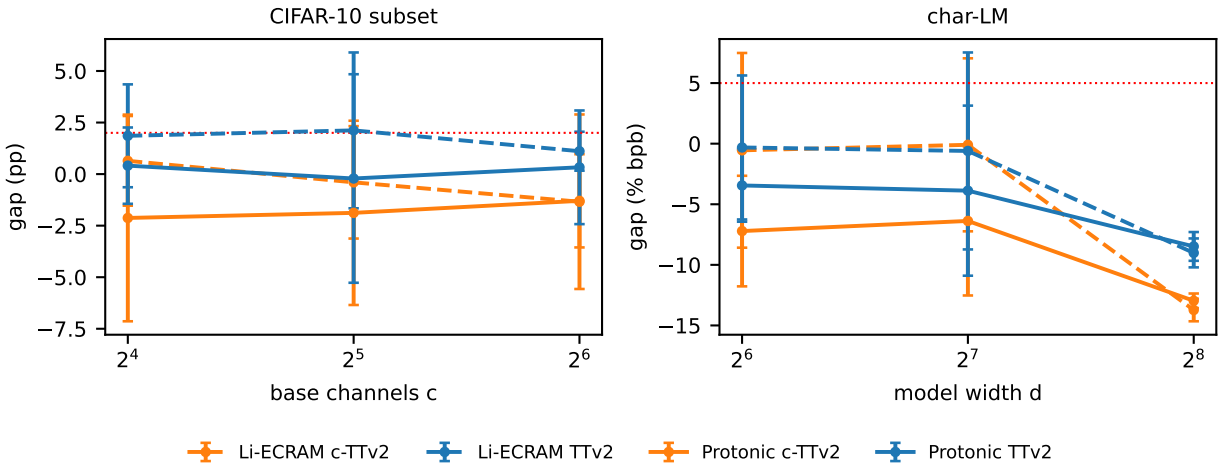


Figure 3: Gap against width with every arm at its own validation-chosen learning rate (mean and 95% interval, 3 seeds); solid: protonic, dashed: Li-ECRAM; dotted: the gap line.

- At width 64 (CIFAR) and 256 (LM) that rate is a poor choice for a Tiki-Taka device. The base protonic device with TTv2 at the FP rate has a gap of 7.32 pp and 16.29%.
- At its own rate (TTv2: 0.33 pp and -8.48%) the gap is gone.
- The device arms chose rates up to 10× above FP's. TTv2's two-array structure filters the gradient like a leaky accumulator, the analogy to momentum drawn by [13], and tolerates larger steps.

Caveats.

- Our FP baseline is plain SGD at a fixed budget. At $d = 256$ its validation-chosen rate ($3e-05$) leaves it at 3.100 bits per byte, worse than at $d = 128$ (2.881). Part of the analog arms' advantage at $d = 256$ is FP-SGD's under-training at this budget, not an analog gain.
- Three widths spanning a factor of four are a small range. Our conclusion is limited to it: we find no evidence that the gap grows with width once learning rates are tuned per arm.

4.4 Sensitivity and the device specification sheet

Design. Around a protonic-like base device we move one parameter at a time while keeping the others at the base:

- the base: 1000 levels, asymmetry 0.05, saturation 0.2, write noise 0.3, no drift, reference error 0.02;
- the parameters moved: levels, asymmetry, nonlinearity (saturation γ), write noise, retention and reference error;
- the algorithm: the pre-registered TTv2, our best validation choice, at the FP learning rate;
- the sizes: three CIFAR and three LM sizes, with SGD at the base sizes.

Fig. 4 shows the curves.

The pre-registered sheet degenerates. The rule asks for the weakest value that keeps the gap under the line. On CIFAR, the base device's own TTv2 gap at the FP rate is already near or above the line: 4.03, 2.39 and 7.32 pp at $c = 16$, 32 and 64. Most CIFAR entries therefore read "not reachable" (Table 6). Two things cause this: the CIFAR baseline's seed noise (Section 4.2) and TTv2's need for its own learning rate at larger widths (Section 4.3). On the LM at $d = 64$ and 128 the base passes and the sheet is well defined.

Degradation thresholds. We therefore add a descriptive reading, labelled as added after the data were seen (`DEVIATIONS-PAPER.md` D1): how far each parameter may move before the gap rises by more than the line over the base device's own gap (Table 7). Its main features are consistent across sizes and tasks:

Reference error is the binding specification. With the symmetry-point reference exact ($\sigma_r = 0$) the TTv2 gap vanishes: -1.4 pp on CIFAR and -0.3% on the LM at the base sizes. Already at $\sigma_r = 0.05$ it reaches 12.3 pp and 17.8%. At every size the grid threshold is $\sigma_r \leq 0.02$ of the half-range; the interpolated crossings lie just above it (Table 7). This is the offset sensitivity that c-TTv2 and AGAD were designed to remove [13]. Their better main-table results (Section 4.2) and our validation (Section 4.1) say that a specification sheet for them would relax this line.

The device must saturate. A constant-step device ($\gamma = 0$) fails badly under TTv2 (30.2 pp; 36.1%), as [13] predicts for algorithms that rely on the device's own decay toward its symmetry point. Up to $\gamma \approx 0.5$ is tolerated.

Asymmetry up to about 0.2 is tolerated. At 0.4 training collapses (39.5 pp).

Levels matter less than expected. TTv2 tolerates a few hundred levels: the threshold is ≥ 500 on CIFAR at $c = 32$ and ≥ 200 on the LM at $d = 128$. With 100 levels the gap is 5.7 pp and 12.3%. This is far below the thousand levels plain SGD needs [10].

Write noise up to one to two steps per pulse is tolerated. Four steps hurts (7.9 pp; 21.8%).

Retention. The drift time constant must be $\geq$ 1e4 training steps on the LM ($d = 128$) and $\geq$ 1e5 on CIFAR ($c = 32$). That is several to tens of times the length of our training runs, which are 2340 and 3000 steps. At 10^3 steps training fails (34.0 pp; 33.7%). For a real array the required time is this number of steps times the update time per step.

Table 6: Device specification sheet (E5) for TTv2: the weakest grid value of each parameter that keeps the gap under the line (2 pp CIFAR; 5% LM) with every other parameter at the protonic base (levels 1000, a 0.05, γ 0.2, c_{2c} 0.3, no drift, σ_r 0.02); in parentheses the interpolated crossing. †: the base itself fails and the value shown is the improvement needed; *: passes over the whole grid; n/r: not reachable on the grid. Last column: the joint specification (every axis at its threshold at once), mean gap over 3 seeds, round 1 or the tightened round 2.

Size	Levels	Asym. a	Nonlin. γ	Noise c_{2c}	Retention τ	Ref. err. σ_r	Joint spec
CIFAR 16	n/r	n/r	n/r	n/r	n/r	$\leq$ 0 (0.0104)†	–
CIFAR 32	$\geq$ 2000 (1488)†	n/r	n/r	$\leq$ 0 (0.17)†	n/r	$\leq$ 0 (0.0179)†	–
CIFAR 64	n/r	n/r	n/r	n/r	n/r	n/r	–
LM 64	$\geq$ 500 (292)	$\leq$ 0.2 (0.231)	$\leq$ 0.2 (0.379)	$\leq$ 1 (1.12)	$\geq$ 1e4 (7e3)	$\leq$ 0.02 (0.0239)	r1: 4.74 pass
LM 128	$\geq$ 1000 (612)	$\leq$ 0.2 (0.21)	$\leq$ 0.2 (0.281)	$\leq$ 0.3 (0.727)	$\geq$ 1e4 (8e3)	$\leq$ 0.02 (0.021)	r1: 2.45 pass
LM 256	n/r	n/r	n/r	n/r	n/r	n/r	–

Table 7: Degradation thresholds (descriptive, added after the one-at-a-time data were seen; DEVIATIONS-PAPER.md D1): the weakest grid value of each parameter whose gap stays within the line (2 pp CIFAR; 5% LM) of the protonic base device's own gap (column 3), all other parameters at the base; interpolated crossing in parentheses; * the whole grid stays within; n/r the nearest grid step already leaves the band.

Alg.	Size	Base gap	Levels	Asym. a	Nonlin. γ	Noise c_{2c}	Retention τ	Ref. err. σ_r
SGD	CIFAR 32	19.80	$\geq$ 1000 (520)	$\leq$ 0.05 (0.0813)	$\leq$ 0.2 (0.266)	$\leq$ 1 (1.3)	$\geq$ 1e4 (7e3)	$\leq$ 0.1 (0.15)
SGD	LM 128	39.85	$\geq$ 200 (143)	$\leq$ 0.4*	$\leq$ 0.2 (0.423)	$\leq$ 4*	$\geq$ 1e4 (3e3)	$\leq$ 0.5*
TTv2	CIFAR 16	4.03	$\geq$ 100*	$\leq$ 0.2 (0.204)	$\leq$ 0.5 (0.606)	$\leq$ 0.3 (0.789)	$\geq$ 1e5 (7e4)	$\leq$ 0.02 (0.0228)
TTv2	CIFAR 32	2.39	$\geq$ 500 (265)	$\leq$ 0.2 (0.209)	$\leq$ 0.5 (0.578)	$\leq$ 2 (2.31)	$\geq$ 1e5 (2e4)	$\leq$ 0.02 (0.0241)
TTv2	CIFAR 64	7.32	$\geq$ 100*	$\leq$ 0.2 (0.203)	$\leq$ 2*	$\leq$ 2 (2.62)	$\geq$ 1e5 (4e4)	$\leq$ 0.02 (0.0244)
TTv2	LM 64	3.18	$\geq$ 200 (130)	$\leq$ 0.2 (0.284)	$\leq$ 0.5 (0.717)	$\leq$ 2 (2.03)	$\geq$ 1e4 (5e3)	$\leq$ 0.02 (0.0326)
TTv2	LM 128	4.25	$\geq$ 200 (167)	$\leq$ 0.2 (0.276)	$\leq$ 0.5 (0.705)	$\leq$ 1 (1.67)	$\geq$ 1e4 (6e3)	$\leq$ 0.02 (0.028)
TTv2	LM 256	16.29	$\geq$ 100*	$\leq$ 0.4*	$\leq$ 2*	$\leq$ 2 (2.64)	$\geq$ 1e4 (4e3)	$\leq$ 0.02 (0.0351)

Plain SGD stays far above the line at every point of the grid on both tasks. No single parameter, moved alone from a protonic-like device, makes plain SGD work.

Pairs and the joint specification. Pairwise, on CIFAR the mean interaction over 60 pairs is -0.2 pp, and 9 of 60 pairs exceed the sum of their single effects by more than the line; on the LM the mean interaction over 60 pairs is 1.5 %, and 6 of 60 pairs exceed the sum of their single effects by more than the line (Table 8). On both tasks the six largest interactions all pair the shortest retention ($\tau = 10^3$ steps) with a second imperfection. A drifting device compounds every other error, so retention must be specified jointly with the rest. Away from fast drift, the pairwise effects are close to additive. The joint specification, with every parameter at its threshold at once (Table 6, last column), by size: CIFAR 16: not defined; CIFAR 32: not defined; CIFAR 64: not defined; LM 128: passes (gap 2.45); LM 256: not defined; LM 64: passes (gap 4.74).

5 Energy with the periphery counted

The unit of comparison. We compare training energy per weight per row. A row is one input vector presented to a weight matrix: one sample of an MLP layer, one patch position of a convolution, or one token of the LM. Every input of the model carries a label (records/ENERGY-SYSTEM.json): SPEC, MEASURED, PROJECTION, SIM (counted in our runs) or ESTIMATE.

Table 8: Pairwise sensitivity (E5): the three largest interactions per task, interaction = gap(a,b) − [gap(a) + gap(b) − gap(base)].

Task	Pair	Gap(a,b)	Singles	Interaction
CIFAR	asym=0.1, tau_steps=1e3	46.64	1.82 + 33.96	13.25
CIFAR	asym=0.2, tau_steps=1e3	40.24	1.92 + 33.96	6.75
CIFAR	gamma=0.5, tau_steps=1e3	41.17	3.55 + 33.96	6.05
LM	c2c=1, tau_steps=1e3	76.94	5.27 + 33.74	42.18
LM	gamma=0.5, tau_steps=1e3	76.63	6.28 + 33.74	40.85
LM	asym=0.1, tau_steps=1e3	73.70	4.09 + 33.74	40.11

The GPU. A digital processor spends six FLOP per weight per row: the forward pass, the input gradient and the weight gradient. For the H100 we use:

- its 700 W board power over its 989 TFLOPS dense BF16 peak (SPEC) [42];
- the MEASURED model-FLOP utilisation of a very large production training run, 38–43% on 8K–16K H100s [43].

This gives 10.5 pJ per weight per row, and 7.1 pJ in the GPU's best case.

The analog array. It spends four kinds of energy:

1. **Forward and backward reads.** Two MACs per weight per row, at the MEASURED efficiency of a complete analog-AI chip with its on-chip periphery: 12.4 TOPS/W chip-sustained [35], with 9.76 TOPS/W for 8-bit MVMs as the low end [36]. The PROJECTION of 84 TOPS/W for an RPU training system [10] is the high end.
2. **The update.** Our simulated pulse count per weight per row (SIM) times the published energy per pulse: about 1 fJ projected for Li-ECRAM [4] and 2.5 fJ for the protonic device [7].
3. **Pulse lines and pulse generation.** Half-select line charging at the device's pulse voltage, using our measured slot occupancy, plus one stochastic bit per line per slot, amortised over an $N \times N$ array. All of these are ESTIMATES and are given ranges.
4. **The Tiki-Taka family's extras.** The A-column reads, one ADC conversion each, and the digital buffer updates.

Result. Table 9 and Fig. 6 give the breakdown.

- **Ratio.** For every published electrochemical device and algorithm, the central GPU-to-analog energy ratio is about 32×. Over the full sensitivity range it is 15–234×.
- **The update is not the bottleneck.** The device pulses cost well under one percent of the total. Thanks to update management the pulse count per weight per row is small: a few hundredths on the LM.
- **The read periphery is.** The forward and backward MVMs take at least 97% of the analog energy.

The femtojoule device energies that dominate the literature's headline numbers are therefore not what limits a training system. The ADC/DAC periphery of the reads is.

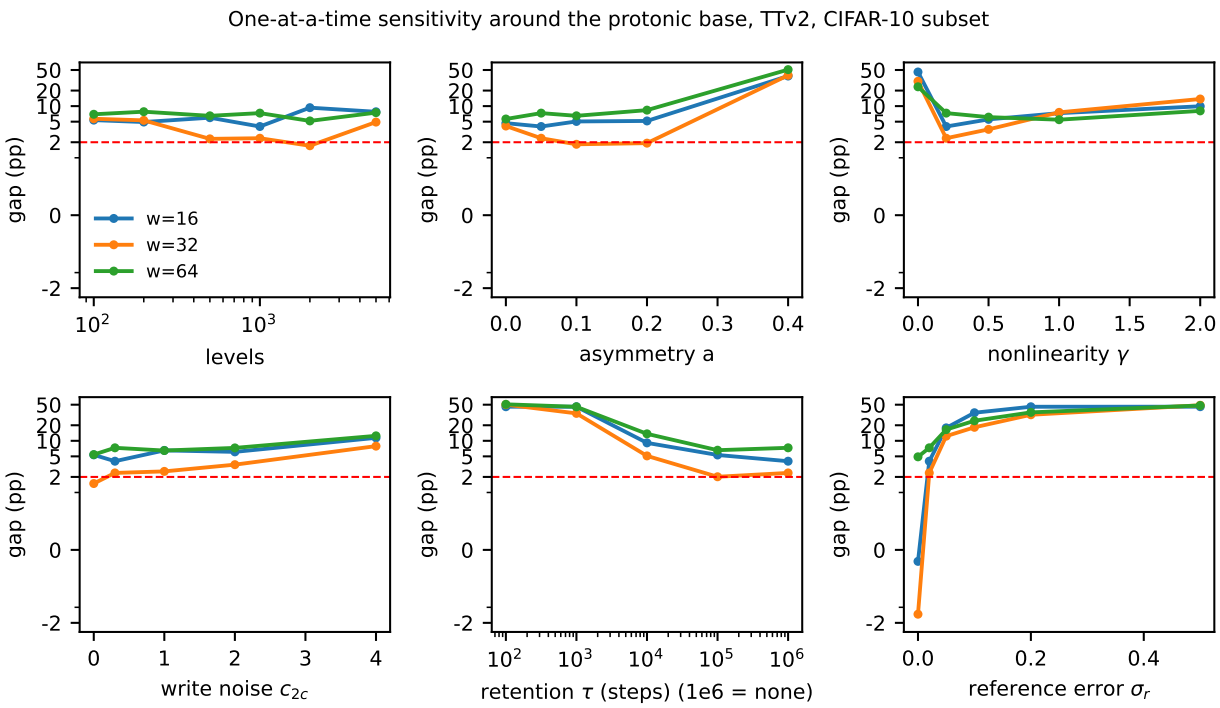


Figure 4: One-at-a-time sensitivity around the protonic base device on the CIFAR-10 subset (TTv2 at the FP learning rate; seed 0, plus seed 1 near the line): gap against each parameter at three widths; dashed: the gap line.

What 100× would need. To reach a 100× advantage over the H100 at its measured utilisation, the reads must reach about 42 TOPS/W including periphery. That is 3.4 times the measured analog-AI chip efficiency used here, and below the RPU projection.

The memistor. Widrow's macroscopic memistor moves about 15 μJ of plating charge per level. Its energy per weight per row is therefore orders of magnitude *above* the GPU's (ratio at most 8.9e-4). This is as expected for wet, centimetre-scale cells: the idea scales only at the nanoscale.

What is measured and what is not.

- MEASURED: the GPU utilisation and the analog MVM efficiency.
- Not measured:
  - the MVM efficiency was measured on phase-change inference chips, not on ECRAM training arrays;
  - ECRAM gate drivers and the three-terminal access devices have no published system-level energy;
  - the update periphery figures are our estimates.

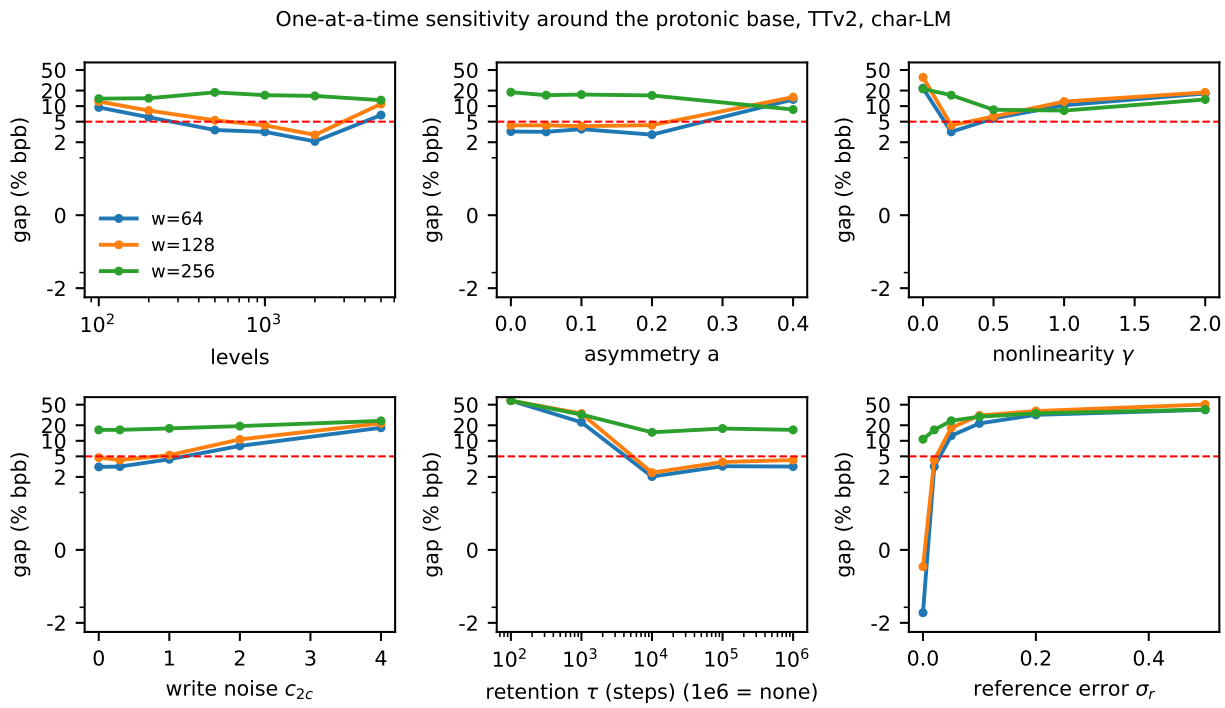


Figure 5: As Fig. 4, for the character LM.

6 Discussion

What the simulations say.

- **The two best published electrochemical specifications can train.** Both the Li-ECRAM fit and the protonic device reach the exact-update baseline within our gap lines, or within the statistical resolution of five seeds, on all three tasks, provided a Tiki-Taka-family algorithm is used.
- **Plain SGD never does,** on any electrochemical device we tried.
- **Width is not the obstacle** we feared. Once each arm has its own learning rate, the gap does not grow with width over the factor of four we could afford.
- **Two device properties decide.** The decisive specifications are not the number of levels, which TTv2 needs only in the hundreds, but:
  - an accurately referenced symmetry point, or an algorithm that does not need one;
  - a saturating (asymmetric) response that gives the gradient accumulator its decay.

  Retention of several training lengths and write noise of up to one or two steps are tolerated.
- **The energy case rests on the periphery,** not on the device. Femtojoule pulses make the update almost free. The forward and backward reads, priced at a measured analog-AI chip's efficiency, set the system at about 32× an H100's energy per weight-row: useful, but not the thousandfold figures that device-only accounting suggests.

Table 9: System energy per weight per row on the character LM (E6), in pJ: forward and backward MVMs at the measured 12.4 TOPS/W of [35], device pulses (simulated count × published or estimated energy per pulse), pulse lines and generation, and the TT-family reads. Ratio = H100 at its measured utilisation over the analog total; the range sets every input at its low or high end (records/ENERGY-SYSTEM.json).

Device	Algorithm	Pulses/w/row	MVM	Device	Lines	TT	Ratio	Range
Li-ECRAM (Tang 2018)	AGAD	0.034	0.323	0.0000	0.0013	0.00003	32.4	16.2–234
Li-ECRAM (Tang 2018)	c-TTv2	0.042	0.323	0.0000	0.0013	0.00003	32.3	16.2–234
Li-ECRAM (Tang 2018)	SGD	0.001	0.323	0.0000	0.0012	0.00000	32.4	16.2–234
Li-ECRAM (Tang 2018)	TTv2	0.038	0.323	0.0000	0.0013	0.00003	32.4	16.2–234
MO-ECRAM (Kim 2019)	AGAD	0.267	0.323	0.0003	0.0015	0.00003	32.3	16.2–232
MO-ECRAM (Kim 2019)	c-TTv2	0.339	0.323	0.0003	0.0015	0.00003	32.3	16.2–232
MO-ECRAM (Kim 2019)	SGD	0.007	0.323	0.0000	0.0013	0.00000	32.4	16.2–234
MO-ECRAM (Kim 2019)	TTv2	0.300	0.323	0.0003	0.0015	0.00003	32.3	16.2–232
Protonic (Onen 2022)	AGAD	0.026	0.323	0.0001	0.0038	0.00003	32.1	15.8–228
Protonic (Onen 2022)	c-TTv2	0.035	0.323	0.0001	0.0038	0.00003	32.1	15.8–228
Protonic (Onen 2022)	SGD	0.001	0.323	0.0000	0.0018	0.00000	32.3	16.1–233
Protonic (Onen 2022)	TTv2	0.033	0.323	0.0001	0.0037	0.00003	32.1	15.8–228
Memistor (Widrow 1960, as claimed)	AGAD	0.017	0.323	262122.1163	0.0012	0.00003	4e-05	2.69e-05–4.26e-05
Memistor (Widrow 1960, as claimed)	c-TTv2	0.022	0.323	323881.6451	0.0012	0.00003	3.24e-05	2.18e-05–3.45e-05
Memistor (Widrow 1960, as claimed)	SGD	0.002	0.323	30015.2982	0.0012	0.00000	0.000349	0.000235–0.000372
Memistor (Widrow 1960, as claimed)	TTv2	0.022	0.323	326250.3670	0.0012	0.00003	3.21e-05	2.16e-05–3.42e-05
W/YSZ/WO_x (Chen 2023)	AGAD	0.007	0.323	0.0000	0.0015	0.00003	32.3	16.2–233
W/YSZ/WO_x (Chen 2023)	c-TTv2	0.010	0.323	0.0000	0.0015	0.00003	32.3	16.2–233
W/YSZ/WO_x (Chen 2023)	SGD	0.000	0.323	0.0000	0.0013	0.00000	32.4	16.2–234
W/YSZ/WO_x (Chen 2023)	TTv2	0.009	0.323	0.0000	0.0015	0.00003	32.3	16.2–233

Widrow's memistor. Taken at its inventor's description (thousands of near-linear levels), the memistor would train as well as modern ECRAM on our tasks. The obstacles of 1960 were the missing multilayer rule, wet macroscopic cells and an energy per level of tens of microjoules. Accuracy was not the obstacle. The idea survived; the implementation had to shrink by about 10 orders of magnitude in energy per update.

6.1 Limitations

- **Simulation only.** Every device is a parametric model.
  - 4 of the 7 catalog entries rest partly or wholly on estimates.
  - The protonic device's write noise and spreads are estimates. A 2026 measurement on BEOL-compatible protonic ECRAM [8] reports 3% cycle-to-cycle and 21% device-to-device variability. These lie inside our tolerated region, but we did not refit the catalog to them.
  - The model has no state-dependent noise beyond the soft bounds, no history effects or relaxation transients beyond a single drift time constant, no temperature, and no endurance limit.
- **Ideal reads.** We isolate the update. Read noise, IR drop, ADC/DAC quantisation and output bounds would add to every gap; our validation omits them and still matches the published numbers. Real arrays also have line resistance and sneak paths. Three-terminal devices avoid the latter but need access transistors.
- **Small models and short budgets.**
  - The networks have 10^5 to 10^6 weights.

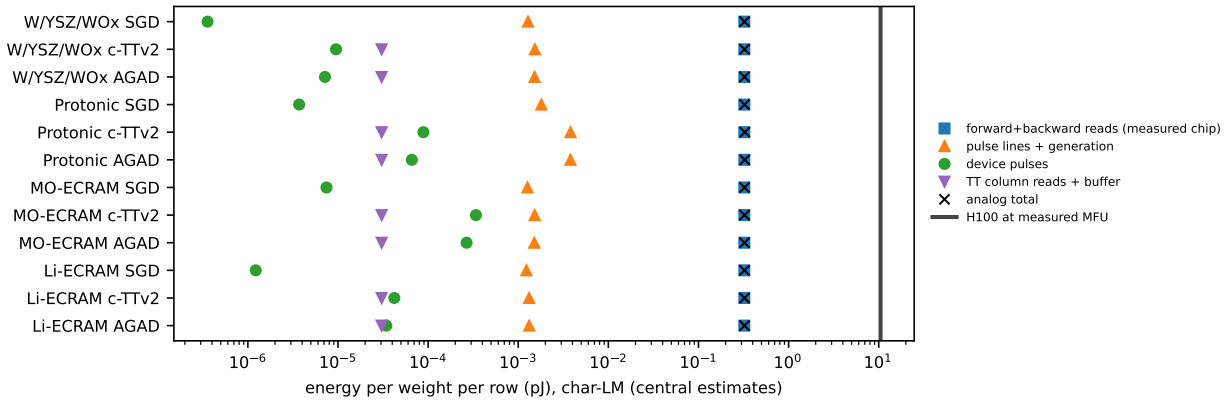


Figure 6: Energy per weight per row on the character LM: the H100 at its measured utilisation against the analog arrays, by component (central estimates; log axis).

  - CIFAR uses a 10,000-image subset whose seed noise (1.41 pp) limits the resolution at the 2 pp line.
  - The LM is a two-layer byte model.
  - The width study spans only a factor of four.

  Larger models are where the remaining questions are. Simulating every pulse becomes expensive; this is also the experience reported by [13].

- **The baseline is SGD.** The FP baseline and every analog arm are plain-SGD-family learners. Against Adam the best analog LM is still about as far behind as FP-SGD is.
- **Mini-batches.** Our update applies a mini-batch's summed pulses as interleaved, rather than sample by sample. The validation shows that batch 16 reproduces published batch-1 results, but strongly asymmetric devices degrade at larger batches.
- **Algorithm details.** We implemented TTv1, TTv2, c-TTv2 and AGAD from the papers' text, not from the authors' code. Every TT-family arm uses a multi-pulse transfer (up to 31 pulses), and a single settings selection on MNIST is shared by all tasks. Results for c-TTv2 and AGAD could improve with per-task tuning.
- **Energy.**
  - The MVM efficiency comes from phase-change inference chips.
  - ECRAM gate drivers and access devices have no published system-level energy.
  - The update periphery is our estimate.
  - The GPU figure uses board power and a published utilisation; it ignores host, network and cooling, which would raise the GPU's energy.
- **Statistics.** Five seeds per cell, no multiplicity correction, and single seeds for most points of the sensitivity grids. Several cells are UNRESOLVED; more seeds, not more conditions, would settle them.

7 A roadmap for groups with device and fabrication resources

The simulations narrow what has to be shown in hardware. We propose four stages. Each has pass criteria taken from this paper's specification sheet (Tables 6 and 7) and main results, so a group can decide quickly whether its device is worth an array.

Stage 1: single-device characterisation (tens of devices, no array).

- **Measurement.** Apply at least 1000 up and 1000 down pulses at the intended pulse width and voltage, and repeat at least three times on at least 32 devices. Fit our device model to the records:
  - step against state (giving $\gamma_{\pm}$);
  - up/down asymmetry a;
  - cycle-to-cycle noise c_{2c} (scatter over step);
  - device-to-device spreads;
  - drift time constant.

  The fitting script and the catalog format are in the release.
- **Pass for TTv2-class training** (others at a protonic-like base):
  - asymmetry ≤ 0.2;
  - saturation $0 < \gamma$ and $\gamma \leq 0.5$;
  - write noise ≤ 2 steps;
  - levels ≥ 500;
  - retention $\tau \geq$ 1e5 updates.

  These are the CIFAR $c = 32$ degradation thresholds. The LM thresholds are similar (Table 7).
- **Endurance.** On the Li-ECRAM and protonic devices, the tolerant algorithms applied at most 0.042 pulses per weight per row on our LM. A training run over 10^{10} rows therefore needs about 4.2e8 pulses per weight, shared by its two devices. The more than 10^9 updates reported for protonic ECRAM in 2026 [8] would cover this.

Stage 2: the symmetry point. This is the specification the simulations single out.

- **Measurement.** Program each device's reference to its symmetry point by alternating pulses, and measure the residual reference error σ_r (in units of the half-range) and its drift over hours and over 10^6 cycles.
- **Pass for TTv2:** $\sigma_r \leq 0.02$, stable over the training time.
- **Otherwise:** show that chopped TTv2 or AGAD tolerates the measured offsets. Our validation reproduces their robustness up to $\sigma_r = 0.5$ on a coarse device (Section 4.1).

Stage 3: small arrays with hardware in the loop.

- **Hardware.** Two 64×64 arrays per layer, the A and C arrays of the Tiki-Taka family.
- **Protocol.**
  1. Run the forward and backward passes digitally on the host from the arrays' read weights.
  2. Apply each update on chip as stochastic pulse coincidences on A, with column reads of A into a host buffer H and threshold pulses into C.
  3. Reset or chop as the algorithm requires.
- **Workloads.** A linear-regression test (a target matrix learned to below 5% of its untrained error, as in our checks), then an MNIST MLP on 8×8 downsampled digits (64 inputs).
- **Prediction first.** Before training, fit the Stage 1 device model, run this paper's simulator with that fit, and record the predicted accuracy and its seed interval. The pass criterion is that the hardware lands inside the predicted interval and within 2 pp of FP. The second condition shows the device works; the first makes the simulator falsifiable.

Stage 4: full arrays and in-memory reads.

- **Hardware.** 256×256 tiles, enough for the 784–256–128–10 MLP with six to eight tiles per array type, then the CIFAR network of this paper.
- **Measurement.** Move the forward and backward MVMs on chip. Measure the system energy per weight per row with the periphery, against this paper's model (Section 5).
- **The energy target** is reads at about 42 TOPS/W including the ADCs, for a 100× advantage over an H100 at its measured utilisation.

The 1960 memistor. Appendix A gives a bench replication with hobby parts and safety notes. It can measure the memistor's missing parameters and place the cell on the same specification sheet. It is a historical and educational experiment: by energy, the memistor cannot compete.

What to report. For any device, report the fields of our catalog with their provenance: levels, $\gamma_{\pm}$, a, c_{2c}, spreads, retention, reference error, pulse energy and voltage. Fitted parameters with their fitting procedure are more useful to the field than headline level counts.

8 Conclusion

In simulation, memistor-class three-terminal electrochemical weights can train small networks to the exact-update baseline, provided the devices saturate, their reference is accurate (or a chopped algorithm removes the need for it) and a Tiki-Taka-family algorithm moves the gradient. Two published specifications, Li-ECRAM and protonic, meet these conditions. Plain SGD does not work on any of them. Under a fair per-arm tuning, the gap does not grow with width over the range we could test. Counting the periphery, the energy advantage over a well-used H100 is about 32× today, rising past 100× only with read periphery several times better than the best measured analog-AI chip. The remaining questions are experimental, and we have tried to make them cheap to ask.

9 Code and data availability

Everything needed to reproduce this paper is released together:

- the pre-registered plan (`PLAN-PAPER.md`) and its deviations (`DEVIATIONS-PAPER.md`);
- the simulator and its checks;
- the device catalog with per-field provenance;
- every run record (one JSON file per training run, with its configuration, metric, pulse counts and hardware);
- the scripts that derive every number, table and figure of this paper from those records (`analyze.py`, `make_figures.py`, `energy/energy_system.py`);
- the citation check (every reference verified against Crossref, the arXiv API or the live URL; `records/CITE-CHECK.json`);
- the bench protocol.

The archive DOI will be added on deposit (Zenodo). Data: MNIST, CIFAR-10 and enwik8 are public and pinned by digest. All text data used for training is human-written (Wikipedia, 2006).

Author contributions and use of AI

Questeria conceived and directed the study. Claude (Anthropic), an AI system, wrote the simulator and its checks, designed and pre-registered the experiments under Questeria's direction, ran them on rented cloud GPUs, analysed the records, verified the references, and drafted the text.

Under the policies of arXiv [44] and of most journals, an AI system cannot be an author or take responsibility for a paper. For any formal submission the arXiv-compliant variant of this paper lists Questeria as the sole author and states Claude's role in this section. All numbers were produced by scripts from the released records.

Acknowledgments

We thank the authors of the open-access papers and of the open-source AIHWKIT whose published device fits and algorithm descriptions made this study possible. No external funding was received. Compute was rented by the author.

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A Bench replication of the 1960 memistor

This appendix is a protocol for others to run. We built nothing and bought nothing. It is meant to measure, with hobby parts, the properties that Widrow's reports leave open (Section 2): levels, up/down symmetry, linearity, write noise, drift and energy per step. The measured cell can then be placed on the same map as the modern devices. The full protocol, an Arduino sketch, a parts list with estimated prices, and an analysis script that turns the log into a device specification for our simulator are in the repository (`BENCH.md`, `bench/`).

Pass criteria. The cell must show at least 32 near-linear, symmetric levels on one lead: over at least 32 consecutive equal-charge pulses in each direction,

1. the mean up step and the mean down step agree within 10%;
2. the step varies by less than $\pm 30\%$ across the range ($\gamma \leq 0.3$);
3. the pulse-to-pulse scatter is below half a step ($c_{2c} \leq 0.5$).

A four-input ADALINE trained by LMS on four such leads should then cut its mean squared error at least tenfold.

Build.

- Electrolyte: 0.5 M copper(II) sulphate in distilled water. Sulphuric acid and brighteners are deliberately left out for safety, so expect some hysteresis.
- Electrodes: 2 mm HB graphite leads with wrapped-wire contacts, the lower contact insulated with nail polish, a wetted length of about 20 mm, and a bare copper counter electrode.
- Electronics: an Arduino Nano, an INA219 current/voltage sensor, and a CD4052 multiplexer for four-wire reads.

Protocol.

1. Read-noise baseline.
2. Plating range in 1 mC pulses.
3. 64 pulses down and 64 up from mid-range, repeated, to measure the step against conductance, the asymmetry and the scatter.
4. A two-hour (ideally overnight) drift run, covered and then stirred, to separate oxygen-driven corrosion.
5. The integral of cell voltage × current per pulse, which gives the energy per step.
6. 20 full cycles to measure endurance.
7. The LMS ADALINE.

Safety.

- Copper sulphate is harmful if swallowed, irritates eyes and skin, and is very toxic to aquatic life. Wear splash goggles and nitrile gloves, and work on a tray away from food, children and pets.
- Use low voltage only (USB). Keep electronics out of splash range, keep aluminium away from the solution, and reduce the current if bubbles form.
- Never pour waste down the drain. Take it to household hazardous-waste collection.
- For eye contact, rinse for 15 minutes and seek medical advice.

What the bench cannot show. A pencil-lead memistor is macroscopic. Its energy per step is set by the copper moved, about 15 μJ per level by Widrow's figures and 986 μJ for our pencil-lead estimate. That is many orders of magnitude above an H100's energy per weight update (Section 5). The bench can confirm or refute the linearity and symmetry the memistor promised; it cannot make training cheaper. The cheap direction, if any, is the nanoscale descendants.

B Hyperparameter selection

Learning rates and algorithm settings were selected on validation splits only, with seed 0.

- **MNIST learning rate.** The FP grid was $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$, extended by the edge rule. The choice is 0.003.
- **TT-family settings.** They were chosen on MNIST validation with a mid-quality device (128 levels, asymmetry 0.1, γ 0.3, c_{2c} 0.5, σ_r 0.02):
  - TTv2: $\lambda \in \{0.1, 0.3, 1, 3\}$ and a period of 16 or 64 batches, with or without a matched A learning rate;
  - c-TTv2: $\lambda \in \{0.1, 0.3\}$, $\rho \in \{0.05, 0.2\}$, with or without reset;
  - AGAD: $\lambda \in \{0.1, 0.3\}$, $\rho \in \{0.05, 0.2\}$, $\beta \in \{0.3, 1\}$.

- **Derived choices.** The better chopped algorithm (c-TTv2) is used in the scale study, and the best of the three (TTv2) in the specification sheet.
- **CIFAR and LM learning rates** were chosen per width (Section 4.3).

Table 10: Settings chosen on MNIST validation with the mid device (Appendix B); validation accuracy in %.

Algorithm	Chosen setting	Val. acc.
TTv2	λ 0.1, A lr $\times$10, period 16	97.86
c-TTv2	λ 0.1, A lr $\times$10, period 16, ρ 0.2, reset yes	97.82
AGAD	λ 0.1, A lr $\times$10, period 16, ρ 0.05, β 0.3, reset no	97.18

C Simulator checks

Each check runs twice. On the real code it must pass. Once for every planted fault it must fail, which shows that the check catches that fault. A crash under a fault also counts as caught. Both suites ran on the desk before the plan was committed, and again at the start of every cloud session; any failure would have stopped the session.

Table 11: Simulator checks and the planted faults each one catches (records/TESTS-PAPER.json, records/TESTS-laneME-checks-on-paper-sim.json).

Check	Result	Planted faults caught
`c_asym_drift`	pass	asymmetry ignored
`c_c2c_noise`	pass	write noise $\sim$ n not sqrt(n)
`c_constant_step_reference`	pass	constant-step reference at a bound (the D1 bug)
`c_drift`	pass	drift relaxes to 0 not bare
`c_fp_equivalence`	pass	apply_exact sign flipped
`c_level_step`	pass	step size halved
`c_pulse_mean`	pass	pulse slots BL dropped; pulse up/down swapped
`c_symmetry_point`	pass	symmetry point sign
`c_ttv2_learns`	pass	TT transfer sign flipped
`c_family_learns`	pass	A-update inputs not chopped
`c_gamma_d2d_sp`	pass	gamma spread shared by the whole tile
`c_identity_sgd_ttv2`	pass	down-branch saturation uses gamma_up
`c_offset_rejection`	pass	AGAD reference never set; chopper demodulation missing; chopper never flips
`c_ttv1_direction`	pass	TTv1 transfer sign flipped

D Earlier development results

This study began as an exploratory screen with the same simulator (SGD and TTv2 only, 1–3 seeds). The screen is summarised here because some of its observations motivated the pre-registered experiments:

- a CIFAR gap "just over 2 pp";
- a gap that "grows with width";

- a “pencil lead built today” that fails every task.

The earlier runs differ from this paper’s in three ways:

- MNIST ran on a different GPU;
- the LM’s selection split overlapped its training data (corrected here, Section 3.5);
- CIFAR and the LM had one or two seeds.

With five seeds the earlier point estimates are mostly confirmed (Table 12), but the earlier pass/fail readings were not meaningful at that sample size. The CIFAR baseline’s seed-to-seed spread is comparable to the gap line itself.

The pessimistic memistor and the pencil-lead estimate were run only in the earlier screen. Both fail every task with SGD and with TTv2. Their specifications are estimates; the bench protocol of Appendix A would replace the pencil-lead estimate by measurements.

Table 12: Earlier development results (lane ME: same simulator, 1–3 seeds, MNIST on a different GPU, the LM selected on an overlapping split) beside this paper’s pre-registered 5-seed values. Gaps in pp (MNIST, CIFAR) and % (LM). The pessimistic memistor and the pencil lead appear only in the earlier runs.

		MNIST		CIFAR		LM	
Device	Alg.	earlier	this paper	earlier	this paper	earlier	this paper
Li-ECRAM (Tang 2018)	SGD	5.24 (n=3)	4.85 (n=5)	21.78 (n=1)	20.81 (n=5)	39.75 (n=1)	45.58 (n=5)
Li-ECRAM (Tang 2018)	TTv2	1.17 (n=3)	1.32 (n=5)	2.15 (n=2)	2.95 (n=5)	4.08 (n=2)	3.52 (n=5)
Protonic (Onen 2022)	SGD	3.67 (n=3)	3.07 (n=5)	19.54 (n=1)	15.77 (n=5)	37.62 (n=1)	40.82 (n=5)
Protonic (Onen 2022)	TTv2	0.77 (n=3)	1.04 (n=5)	2.13 (n=2)	2.13 (n=5)	4.21 (n=2)	3.64 (n=5)
MO-ECRAM (Kim 2019)	SGD	7.20 (n=3)	6.96 (n=5)	26.19 (n=1)	25.29 (n=5)	76.49 (n=1)	50.84 (n=5)
MO-ECRAM (Kim 2019)	TTv2	1.84 (n=3)	1.80 (n=5)	8.04 (n=1)	5.45 (n=5)	17.19 (n=1)	16.51 (n=5)
W/YSZ/WO_x (Chen 2023)	SGD	8.19 (n=3)	8.25 (n=5)	37.88 (n=1)	36.82 (n=5)	54.79 (n=1)	50.67 (n=5)
W/YSZ/WO_x (Chen 2023)	TTv2	1.49 (n=3)	1.48 (n=5)	10.45 (n=1)	10.59 (n=5)	20.77 (n=1)	21.17 (n=5)
Memistor (Widrow 1960, as claimed)	SGD	1.71 (n=3)	1.29 (n=5)	17.15 (n=1)	9.86 (n=5)	28.01 (n=1)	30.94 (n=5)
Memistor (Widrow 1960, as claimed)	TTv2	0.63 (n=3)	0.73 (n=5)	3.24 (n=1)	2.86 (n=5)	10.75 (n=1)	8.99 (n=5)
Memistor (pessimistic)	SGD	32.58 (n=3)	–	41.52 (n=1)	–	47.26 (n=1)	–
Memistor (pessimistic)	TTv2	15.33 (n=3)	–	39.16 (n=1)	–	27.44 (n=1)	–
Pencil lead built today	SGD	23.05 (n=3)	–	43.92 (n=1)	–	42.36 (n=1)	–
Pencil lead built today	TTv2	11.72 (n=3)	–	37.39 (n=1)	–	21.98 (n=1)	–